Department of Computer Science and Engineering
Academic Year 2021 – 2022 (Odd Semester)

Degree, Semester & Branch: I Semester B.E. ECE-'B'
Course Code & Title: GE3151 Problem solving and python programming
Name of the Faculty member (s): Mrs.S.Manjula

Innovative Practice Description

Unit / Topic: Unit I / Illustrative problem: Towers of Hanoi

Course Outcome: CO 1

Topic Learning Outcome: TLO 1,2,3

Activity Chosen: Role Play

Justification:

- Students are asked to "act out" so they get a better idea of the concepts and theories being discussed**.** Role play encourages participation among passive learners, adds dynamism to the classroom and promotes the retention of material. Towers of Hanoi is a kind of game where the students are involved to play the game as per the rule.
- **Time Allotted for the Activity:** 10 minutes

Details of the Implementation:

- Instructor explained the particular concepts/topic in classroom within 35 minutes.
- Three students were asked to come in front of the classroom.
- Student 1 was asked to hold 3 boxes of different sizes.
- Student 2 was acted as auxiliary.
- The objective of the puzzle game is to move the entire stack to another rod, obeying the following simple rules:
  - Only one disk can be moved at a time.
  - Each move consists of taking the upper disk from one of the stacks and placing it on top of another stack i.e. a disk can only be moved if it is the uppermost disk on a stack.
  - No disk may be placed on top of a smaller disk.
- Students were asked to follow the game rules and finally student 3 should hold all the boxes by interacting with other students with optimal path cost of 2^N-1.
- Students enthusiastically play the game and they completed the game correctly in 2nd time.
- This assisted the students in recalling the topic taught on that specific day, generating new ideas about the topic, and answering questions about the topic with ease.

CO – PO / PSO mapping:

CO	PO1	PO2	PO3	PSO1
CO 1	2	1	1	1

(1 – Low 2 – Moderate 3 – High)

PO / PSO mapped:

Innovative practice	**PO1**	**PO2**	**PO3**	**PSO1**
	2	1	1	1
Justification for correlation	Apply basics mathematics knowledge to solve the problem	Solutions can be identified, formulated to solve the problem	Solution for the problem can be drawn using flow chart	Helped the students in solving various problems in real world

- **Images / Screenshot of the practice:**

- **Reflective Critique:**
    - ***Feedback of practice from students and other stakeholders:***
      Through this activity the students felt easy to write algorithm and draw flow chart.
    - ***Benefit of the practice:***
        - From this activity, the students can get more clarity in the particular topic.
        - Students can able to explain the concepts without any confusion.
        - This activity brought out an interest in them and provided a happy & joyful learning environment.

- This practice has made the class more interactive than the conventional practice

Challenges faced in implementation:

- This activity involves only 3 members.
- Not able to make all the students to involve in this activity

References:

- https://serc.carleton.edu/introgeo/roleplaying/whatis.html
- https://www.ritrjpm.ac.in/images/computer-science/10_CS8591_Rolepaly.pdf

Signature of Faculty Member **HOD**